

Supplementary Material

Task-Oriented Dynamic Modeling and Aggregation of AI Data Center Plants

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This document specifies the supply-chain equations, electrical parameters, workload construction, aggregation rules and additional validation of the accompanying paper. The electrical equations follow Zhao and Zhao [1]. Task timing follows Ko and Zhu [2], with power levels from the National Laboratory of the Rockies (NLR) workload dataset [3]. The new material concerns their plant-level composition and the evaluation of physical equivalents. All numerical results in this document come from saved project results. No additional simulations are introduced by this document.

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S1 Electrical Model and Power Accounting

S1.1 Conventions and state variables

A reproducible plant equivalent requires consistent device bases and connection signs. A chain contains an active front end (AFE), a voltage source inverter (VSI), a power supply unit (PSU) and a downstream DC–DC stage. Each supply chain uses its own three-phase rating S_n as the power base. The network uses $S_b = 200$ MVA and $\omega_b = 2\pi 60$ rad/s. Time is in seconds. Per-unit inductances and capacitances multiply derivatives divided by ω_b . Controller integrators retain the unscaled time derivatives used in the source equations. The reference speed is $\omega_s = 1$.

The local AFE frame rotates by the phase-locked loop (PLL) angle θ . Define

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \quad (\text{S1.1})$$

Here $\mathbf{v}_g$ is the voltage at the low-voltage terminal of the unit transformer in network coordinates. The rectifier current $\mathbf{i}_a$ is positive into the supply chain. The VSI uses a separate synchronous frame with voltage $\mathbf{v}_v$ and converter current $\mathbf{i}_c$.

One chain has 21 differential states. The AFE contributes θ , ϵ , $\bar{v}_q$, the two entries of $\mathbf{i}_a$, ξ_a and the two entries of $\boldsymbol{\gamma}_a$. The VSI contributes the two entries of each of $\mathbf{i}_c$, $\mathbf{v}_v$, $\boldsymbol{\xi}_v$ and $\boldsymbol{\gamma}_v$. The remaining states are the shared DC-link voltage v_d , the PSU voltage v_p and integrator ξ_p , and the server-side voltage v_e and integrator ξ_e . The following vector equations count each two-entry vector as two states.

S1.2 AFE rectifier

The rectifier synchronizes to the unit-transformer voltage and regulates the shared DC link. Let $\mathbf{v}_a = \mathbf{R}(\theta)\mathbf{v}_g$. Its equations are

$$\dot{\theta} = \omega_b(\omega_{\text{pll}} - \omega_s), \quad \dot{\epsilon} = \bar{v}_q, \quad (\text{S1.2})$$

$$\dot{\bar{v}}_q = \omega_{\text{lp}}(v_{a,q} - \bar{v}_q), \quad \omega_{\text{pll}} = \omega_s + k_{p,\text{pll}}\bar{v}_q + k_{i,\text{pll}}\epsilon, \quad (\text{S1.3})$$

$$\frac{\ell_a}{\omega_b}\dot{\mathbf{i}}_a = \mathbf{v}_a - v_d\mathbf{m}_a - r_a\mathbf{i}_a + \omega_{\text{pll}}\ell_a\mathbf{J}\mathbf{i}_a, \quad (\text{S1.4})$$

$$\dot{\xi}_a = v_d^* - v_d, \quad \dot{\boldsymbol{\gamma}}_a = \mathbf{i}_a - \mathbf{i}_a^*, \quad (\text{S1.5})$$

$$\mathbf{i}_a^* = \begin{bmatrix} k_{p,da}(v_d^* - v_d) + k_{i,da}\xi_a \\ 0 \end{bmatrix}, \quad (\text{S1.6})$$

$$\mathbf{m}_a = \frac{k_{p,ca}(\mathbf{i}_a - \mathbf{i}_a^*) + k_{i,ca}\boldsymbol{\gamma}_a + \omega_{\text{pll}}\ell_a\mathbf{J}\mathbf{i}_a}{v_d}. \quad (\text{S1.7})$$

A superscript $\star$ denotes a reference value. The constants r_a and ℓ_a are the AFE filter resistance and inductance. Subscripts da and ca identify the DC-voltage and current controllers. The current-controller error and the coupling signs follow the source model [1]; replacing them by another rectifier convention changes the model.

S1.3 VSI and shared DC link

The VSI regulates its output voltage while drawing energy from the rectifier DC link. With reference $\mathbf{v}_v^* = [v_u^*, 0]^\top$, its equations are

$$\frac{\ell_v}{\omega_b}\dot{\mathbf{i}}_c = v_d\mathbf{m}_v - \mathbf{v}_v - r_v\mathbf{i}_c + \omega_s\ell_v\mathbf{J}\mathbf{i}_c, \quad (\text{S1.8})$$

$$\frac{c_v}{\omega_b}\dot{\mathbf{v}}_v = \mathbf{i}_c - \mathbf{i}_o + \omega_s c_v \mathbf{J} \mathbf{v}_v, \quad (\text{S1.9})$$

$$\dot{\boldsymbol{\xi}}_v = \mathbf{v}_v^* - \mathbf{v}_v, \quad \dot{\boldsymbol{\gamma}}_v = \mathbf{i}_c^* - \mathbf{i}_c, \quad (\text{S1.10})$$

$$\mathbf{i}_c^* = k_{p,vv}(\mathbf{v}_v^* - \mathbf{v}_v) + k_{i,vv}\boldsymbol{\xi}_v - \omega_s c_v \mathbf{J} \mathbf{v}_v, \quad (\text{S1.11})$$

$$\mathbf{m}_v = \frac{k_{p,cv}(\mathbf{i}_c^* - \mathbf{i}_c) + k_{i,cv}\boldsymbol{\gamma}_v - \omega_s\ell_v\mathbf{J}\mathbf{i}_c}{v_d}, \quad (\text{S1.12})$$

$$\frac{c_d}{\omega_b} \dot{v}_d = \mathbf{m}_a^T \mathbf{i}_a - \mathbf{m}_v^T \mathbf{i}_c. \quad (\text{S1.13})$$

Here r_v , ℓ_v and c_v are the VSI filter parameters, c_d is the DC-link capacitance, and $\mathbf{i}_o$ is the current drawn by the PSU array. Subscripts vv and cv denote its voltage and current controllers. The DC-link equation couples both converters through their current balance. The study represents online double-conversion operation. Battery discharge, bypass switching and protection actions are outside this model.

S1.4 PSU array and server DC–DC stage

The positive-sequence PSU equivalent converts its AC input into a common DC supply. Let g_p be its controlled input conductance and i_p its DC output current. Then

$$\frac{c_p}{\omega_b} \dot{v}_p = \frac{(g_p - r_p g_p^2) \mathbf{v}_v^T \mathbf{v}_v}{3v_p} - i_p, \quad \dot{\xi}_p = v_p^* - v_p, \quad (\text{S1.14})$$

$$g_p = k_{p,vp}(v_p^* - v_p) + k_{i,vp}\xi_p, \quad \mathbf{i}_o = g_p \mathbf{v}_v. \quad (\text{S1.15})$$

The parameters r_p and c_p represent the PSU equivalent resistance and capacitance. The downstream DC–DC stage has a fast current loop represented by its commanded current i_e , with

$$\frac{c_e}{\omega_b} \dot{v}_e = i_e - g_e v_e, \quad \dot{\xi}_e = v_e^* - v_e, \quad (\text{S1.16})$$

$$g_e = \frac{p_{\text{load}}}{3(v_e^*)^2}, \quad i_e = k_{p,ve}(v_e^* - v_e) + k_{i,ve}\xi_e, \quad i_p = \frac{v_e}{v_p} i_e. \quad (\text{S1.17})$$

The factor 3 follows the source model's three-phase power convention. The input p_{load} is the prescribed server demand on the chain base at $v_e = v_e^*$. The instantaneous power of the modeled load is $3g_e v_e^2$, so it equals p_{load} at the voltage reference. This distinction matters when interpreting internal voltage excursions. The node-power waveform specifies the terminal-demand command; it is not an additional current injection at the PCC.

S1.5 Hall and network connections

Each hall has one common feeder and four parallel supply branches. Each branch has its own 3 MVA unit transformer and independent converter output. The four branches share one quarter of the hall demand command and have identical parameters within a hall. The cooling load connects to the hall bus after the common feeder and before the unit transformers. Auxiliary loads connect to the two medium-voltage buses. The buses and both main transformers remain distinct in every equivalent.

The current drawn from the network by chain n is

$$\mathbf{i}_{n,\text{sys}} = \frac{S_n}{S_b} \mathbf{R}(\theta_n)^T \mathbf{i}_{a,n}. \quad (\text{S1.18})$$

With the network voltage phasors V , generator injections I_G and explicit load withdrawals I_L , nodal balance is $YV = I_G - I_L$. A load represented as an admittance is included in Y and is excluded from I_L . The algebraic network retains line and transformer impedances and feeder charging. It omits their electromagnetic differential states.

For the full plant, the operating-point cooling power is 17% of the hall's AC IT input. Total auxiliary power is 3% of the plant's AC IT input, divided equally between the medium-voltage sections. Cooling uses a constant-power representation and the auxiliaries use constant impedance. The static demand at bus 8 is

$$P_{8,\text{residual}} = 100 - 1.20 \sum_n P_{n,\text{AC}} \text{ MW}. \quad (\text{S1.19})$$

The original 35 Mvar remains as a static impedance load. Collector losses are accounted for by the network. The PCC power is the sum of the powers entering both main transformers, so it includes these losses. The full plant's static loads are retained in its equivalents. They are not adjusted to cancel an equivalent's loss error.

The external system has a fourth-order synchronous machine with excitation and turbine-governor dynamics at bus 1, a droop grid-forming converter at bus 2 and a phase-locked-loop grid-following converter at bus 3. Their equations and parameter conventions follow the modified nine-bus reference in the supplement of [1]. The implementation uses the source-current feed-forward sign -1 and the q-axis voltage convention for the grid-following converter. The accompanying data files give all grid parameter values and network branches. These grid devices contribute 37 states, so the plant model has $37 + 21K$ states after aggregation. The full plant has $K = 48$ and 1045 states.

S2 Electrical Parameters and Workload Inputs

S2.1 Device design

Independent parameter draws describe differences between halls while preserving identical chains inside each hall. A PCG64 generator uses seed 20260919. The draw order is section, hall, the nine passive elements, seven controller bandwidths, the PLL filter frequency, feeder length and unit-transformer impedance. The numerical ranges are study assumptions, rather than a measured distribution of data-center equipment.

Table S1: Nominal Device Parameters and Uniform Draw Ranges

Quantity	Symbol	Nominal	Draw range
AFE resistance	r_a	0.003	0.0027 to 0.0033
AFE inductance	ℓ_a	0.05	0.045 to 0.055
DC-link capacitance	c_d	2.0	1.8 to 2.2
VSI resistance	r_v	0.003	0.0027 to 0.0033
VSI inductance	ℓ_v	0.05	0.045 to 0.055
VSI capacitance	c_v	0.2	0.18 to 0.22
PSU resistance	r_p	0.005	0.0045 to 0.0055
PSU capacitance	c_p	2.0	1.8 to 2.2
Server-side capacitance	c_e	0.2	0.18 to 0.22

All entries are per unit on the chain base. The voltage references are $v_d^* = v_u^* = v_p^* = 1$ and $v_e^* = 0.5$. Nominal values originate from [1]. The VSI voltage-loop bandwidth is 80 Hz in this study, whereas the source Table I uses 100 Hz.

Table S2: Controller Bandwidths and Tuning Damping Ratios

Loop	Nominal in Hz	Range in Hz	ζ
PLL	20	16 to 24	0.707
AFE DC voltage	5	4 to 6	1
AFE current	200	160 to 240	0.707
VSI voltage	80	64 to 96	1
VSI current	400	320 to 480	1
PSU voltage	10	8 to 12	1
Server voltage	100	80 to 120	1

The PLL low-pass frequency is uniform from 80 to 120 Hz. Gains follow the bandwidth rule in [1]. With $\omega_n = 2\pi f_{bw}$, the voltage loop has $k_p = 2\zeta\omega_n C/\omega_b$ and $k_i = \omega_n^2 C/\omega_b$. The current loop has $k_p = 2\zeta\omega_n L/\omega_b - R$ and $k_i = \omega_n^2 L/\omega_b$. The PLL uses $k_p = 2\zeta\omega_n/\omega_b$ and $k_i = \omega_n^2/\omega_b$. Gains are derived from each draw and are not independent random parameters.

Exact values for all 12 halls, including every derived gain, are in `data/design.json`. Table S3 gives the controller and connection parameters that identify the hall differences. The data file retains full precision.

Table S3: Hall Design Summary. Frequencies in Hz, Feeder Length in km, Transformer Impedance in Percent.

Hall	PLL	AFE voltage	VSI voltage	PSU voltage	Feeder	Transformer
1	19.17	4.22	77.58	8.14	0.92	5.93
2	18.67	5.39	91.66	10.52	0.95	5.74
3	18.25	5.47	71.17	10.50	0.65	5.26
4	23.56	5.52	72.54	10.90	1.25	5.30
5	17.39	5.88	73.97	9.48	1.49	5.77
6	17.92	5.54	71.70	10.34	1.23	5.01
7	17.93	4.85	64.84	8.17	1.01	5.21
8	22.56	4.30	94.19	10.87	1.28	5.42
9	20.33	5.97	86.54	9.54	0.22	5.54
10	23.18	5.88	94.08	9.07	0.83	5.37
11	19.71	4.45	95.56	10.59	0.94	5.12
12	21.27	5.23	73.92	10.07	0.94	5.54

S2.2 Network data

The main transformers are each 100 MVA with 8.5% impedance and assumed $X/R = 35$. Unit transformers are 3 MVA with a uniformly drawn impedance of 5% to 6% and assumed $X/R = 6$. Their ratings are compatible with the repeated power-block representation discussed in [4]; the complete topology is a study configuration.

Each 34.5 kV feeder has a drawn length from 0.2 to 1.5 km. Its series impedance is $0.046 + i0.043 \Omega$ per 1000 ft, with capacitive reactance coefficient $0.042 \text{ M}\Omega \cdot 1000 \text{ ft}$. These values come from Southwire SPEC 80214, revision 1.000.021, for a 35 kV, 500 kcmil aluminium cable. The admittance uses the line length and both charging halves. Transformer and feeder impedances are converted to the system base before forming Y .

The three external buses with loads have 125 MW and 50 Mvar at bus 5, 90 MW and 30 Mvar at bus 6, and the 100 MW and 35 Mvar budget at bus 8. Bus 1 is the slack source. The active-power schedules at buses 2 and 3 are 163 and 85 MW. Voltage set points are 1.04, 1.025 and 1.025 p.u. at buses 1 to 3. The branch table in `data/grid_parameters.json` retains its 100 MVA source base and the explicit conversion to 200 MVA.

S2.3 Power measurements and constructed task timing

Power levels and timing have separate sources. The NLR measurements sum graphics processing unit (GPU) board power and central processing unit (CPU) package power on nodes with four H100 GPUs and two AMD EPYC 9554 processors [3]. They exclude memory, network cards, fans, storage and power-supply losses. The training label denotes Stable Diffusion, fine-tuning denotes Llama-2 70B LoRA, and online inference denotes Llama-3.1 70B serving. The training and fine-tuning records use 0.2 s polling and serving uses 0.1 s polling. These intervals do not establish sensor bandwidth.

The phase model supplies the fast timing. The base frequency is drawn once per task from 0.5 to 1.5 Hz for training or 0.3 to 0.7 Hz for fine-tuning. Each iteration has duration $[f_j(1 + \xi_i)]^{-1}$ with a zero-mean normal ξ_i of standard deviation 0.1. Compute shares are uniform from 0.55 to 0.80 for training and 0.70 to 0.90 for fine-tuning. Compute-phase level offsets have standard deviations 0.05 and 0.03, respectively. Communication-phase drops have means 0.3 and 0.8 with the same respective standard deviations. These are the experimental settings of [2], not universal workload limits.

Within-phase standard deviations are drawn once per task. Training uses 0.02 to 0.05 in computation and 0.01 to 0.03 in communication. Fine-tuning uses 0.01 to 0.03 and 0.005 to 0.02. Independent standard normal samples are generated before integration on a 0.01 s grid. Linear interpolation between these samples fixes the temporal dependence for every solver evaluation. This interpolation is a study assumption. It does not turn the NLR records into measured fast task cycles.

The compute power is the median above the midpoint of the fifth and ninety-fifth percentiles of the selected measured window. Each node’s constructed power is restricted at all input knots to the interval

between 418.2 W and the maximum of that measured window. Phase boundaries retain distinct left and right values. Linear interpolation is applied only inside each phase. All nodes assigned to one job use the same realization. Independent jobs use different seeded realizations.

Online inference uses a saturating fit of power against request rate from the two measured serving sweeps. Its request rate remains constant in each simulation window. Separate requests across replicas do not supply evidence for a common high-frequency inference fluctuation. The exact fit and its source intervals are retained in `data/inference_map.json`.

S2.4 Job sizes, allocation and independent inputs

Every hall has 2800 node slots. The baseline occupancy is drawn uniformly from 0.55 to 0.95 using seed components 20260919 and 1. One eighth of the baseline training or fine-tuning slots are retained for synchronous jobs, rounded down within each hall. Released slots serve online inference. Idle slots retain the measured idle-node power. This construction preserves the occupied resource count without adding the GPU or CPU power a second time.

The synchronous job size follows a registered screen of the full model. The candidate sequence is 1, 0.5, 0.25 and 0.125 of the baseline job slots. The selected value is 0.125. Selection uses the full plant's response and converter reference levels before equivalent-model errors are evaluated. It does not establish a maximum safe job size for a physical facility. The supply-chain model contains no active device current, modulation or protection limits.

S1 has one training job across all halls. S2 has six independently generated training jobs, one per hall pair. S3 combines training and fine-tuning with hall-specific resource shares. Every scenario also contains online serving. Each task is scaled once by its total node count. Hall shares are $a_{hj} = m_{hj} / \sum_h m_{hj}$, so their sum is one. All inter-hall time offsets are zero. The grouping and test inputs use separate random draws and the corresponding source windows. The exact seeds, node counts, phase tables, source record identifiers and allocation matrices are stored in `data/scenarios.json`.

S3 Aggregation and Validation Definitions

S3.1 Feature construction

A consistent comparison requires both feature sets to use the same grouping inputs and equivalent construction. The 19 design features are the nine passive elements, seven bandwidths, PLL filter frequency, feeder length and unit-transformer impedance. Mean AC loading supplies the operating-point feature. Each feature is standardized by its common mean and standard deviation over all halls and the three grouping operating points. The fixed feature vector uses the 19 design features once and the three normalized loading values divided by $\sqrt{3}$.

For response features, $G_{s,mh}(i\omega_k)$ is the full model's response from 1 MW at hall h to output m , at grouping operating point s . The four outputs are PCC active power, reactive power and voltage, and synchronous-machine frequency. The 601 logarithmic samples cover 0.1 to 20 Hz. Define $w_k = 1/(2N_b)$ within band b , with the bands 0.1 to 3 Hz and greater than 3 to 20 Hz. The common output scale is

$$\sigma_m^2 = \frac{1}{3H} \sum_{s=1}^3 \sum_{h=1}^H \sum_k w_k |G_{s,mh}(i\omega_k)|^2. \quad (\text{S3.1})$$

For each hall, the real and imaginary parts of $G_{s,mh}\sqrt{w_k}/4/\sigma_m$ are concatenated over frequency and output. The three scenario blocks are divided by $\sqrt{3}$ and concatenated. This retains relative gain and phase differences without allowing one output's unit to set the distance. Test inputs do not enter these scales.

K-means uses Lloyd's algorithm, 20 initializations and random seed 20260919. Each medium-voltage section forms $K/2$ groups. Cross-section merges are prohibited. Tested total group counts are 2, 4, 6, 8, 10 and 12. Fixed and per-scenario memberships coincide for this electrical design. The exact memberships and feature scales are in `data/groups.json`.

S3.2 Physical equivalent

Let a group contain units with ratings S_n and weights $w_n = S_n / \sum_n S_n$. Its rating is $S_E = \sum_n S_n$ and its terminal demand is the sum of the member demands. A capacitance, PI gain or filter cut-off a uses $a_E = \sum_n w_n a_n$. A series resistance or inductance uses

$$a_E = \left(\sum_n \frac{w_n}{a_n} \right)^{-1}. \quad (\text{S3.2})$$

These rules act on device per-unit values referred to the member ratings. For the network branches on the common system base, the equal-loss approximation is

$$Z_E = \frac{\sum_n P_n^2 Z_n}{(\sum_n P_n)^2}. \quad (\text{S3.3})$$

The weights P_n are the full model's initial active flows. Feeder flows include cooling, whereas unit-transformer flows include only the corresponding IT chain. Feeder charging is summed. The approximation does not guarantee an exact loss match when voltage magnitudes or power factors differ. Its PCC power offset is retained and reported.

Four identical chains in one hall reduce exactly to a chain on their summed rating under equal inputs and consistent initial states. The common feeder is unchanged and the four unit transformers form their exact parallel equivalent. Cross-hall aggregation is approximate because device parameters and individual task disturbances differ. The main transformers and both medium-voltage buses are retained.

S3.3 Small-signal and modal quantities

Linearization of $\dot{x} = f(x, z, u)$ and $0 = g(x, z)$ gives a state matrix after removal of the common angle reference and elimination of the nonsingular algebraic block. With derivatives at the operating point,

$$A = f_x - f_z g_z^{-1} g_x, \quad B = f_u - f_z g_z^{-1} g_u, \quad G(s) = C(sI - A)^{-1} B + D. \quad (\text{S3.4})$$

The output derivatives include the corresponding elimination of z . Input enters the device equations through the terminal-demand command. The task response is $G(s)B_c$.

Repeated modes use an invariant-subspace projector $\Pi_c = V_c(W_c^H V_c)^{-1} W_c^H$. For a repeated eigenvalue λ_c , the cluster residue is $C \Pi_c B$. Its task-to-output residue follows after multiplication by B_c . The contribution measure is the residue magnitude divided by $|\text{Re } \lambda_c|$. For the modal scatter plot, marker area uses the largest contribution over all tasks of that model. The highlighted job's transfer curve is evaluated separately. This avoids interpreting the scatter as a job-specific mode filter.

The main comparison uses the worst task's normalized complex response error in a band $\mathcal{B}$,

$$\epsilon_{\mathcal{B}} = \max_j \left[\frac{\sum_{k \in \mathcal{B}} |G_{E,j}(i\omega_k) - G_{F,j}(i\omega_k)|^2}{\sum_{k \in \mathcal{B}} |G_{F,j}(i\omega_k)|^2} \right]^{1/2}. \quad (\text{S3.5})$$

The sums weight the saved logarithmic grid points equally. The worst task is selected separately for each output and model. The main 5% criterion applies to PCC active power in the upper band at both grouping and test operating points. It is a study criterion and does not certify the other outputs. Peak-gain and peak-frequency deviations use each model's refined peak. Phase difference uses the full model's peak grid frequency for both models. These distinct definitions explain why matching a peak gain can coexist with a large complex band error.

S3.4 Time integration and timing

Every model receives the same stored input realization. A one-second rest precedes the 60-second task window. Every phase transition appears in the scheduled-event calendar. The integrator stops at the event,

installs the next continuous segment, solves consistent algebraic initial values and resumes. A finite-width ramp is not substituted for a phase transition.

The production calculation uses Solverz Rodas with KLU, JIT-compiled modules, relative tolerance 10^{-4} , absolute tolerance 10^{-6} , maximum step 0.01 s and output spacing 0.0025 s. Consistent initialization uses tolerance 10^{-6} . Model checks require equal equation and variable counts and initial residual below 10^{-10} . Strict comparisons use relative tolerance 10^{-8} and absolute tolerance 10^{-10} . The exact 12-group control also uses a paired rerun at relative tolerance 10^{-6} to resolve errors near the production tolerance.

For output y , set $d_F(t) = y_F(t) - y_F(0)$ and $d_E(t) = y_E(t) - y_E(0)$. Over the task window, the dynamic error is $\text{RMS}(d_E - d_F)/\text{RMS}(d_F)$. Initial offset $y_E(0) - y_F(0)$, absolute RMS error and maximum dynamic error are separate quantities. Raw plots retain the initial offsets. Internal voltage comparisons map a full-model chain to the equivalent containing that chain's hall; they assess a local-state approximation, not an additional PCC measurement.

Integration timing runs serially. Each model is built once and warmed up once. Three subsequent rounds rotate the model order and report the median and range. The recording function retains PCC active power. Model building and response calculations are excluded from these integration times. These are implementation-specific costs rather than estimates from state count.

S4 Additional Results and Applicability

S4.1 Complete group-count comparison

A method comparison needs both favorable and unfavorable cases. Table S4 gives every tested group count for the frozen electrical design, rather than only the selected equivalents. The response features need 4, 8 and 4 groups for S1, S2 and S3 to keep the upper-band PCC active-power error below 5% at both inputs. The parameter features need 4, 12 and 6 groups. These are the smallest passing values among the tested even group counts.

The coarse two-group equivalent gives a 32.37% upper-band error in the S2 grouping case. Parameter grouping in S2 gives 14.40% at four groups and 16.78% at six groups. Increasing the group count therefore does not ensure decreasing error. In S1, parameter features give smaller band errors than response features at eight and ten groups. Both features use the same physical equivalent construction, so these comparisons concern the grouping rather than different equivalent formulas.

Table S4: Worst-Task PCC Active-Power Band Errors in Percent over 3–20 Hz

Scenario	K	Parameter features		Response features	
		Grouping	Test	Grouping	Test
S1	2	6.28	6.18	6.28	6.18
S1	4	4.88	4.79	2.96	2.90
S1	6	2.09	2.04	1.45	1.41
S1	8	0.60	0.58	1.27	1.26
S1	10	0.28	0.27	0.46	0.45
S1	12	0.00	0.00	0.00	0.00
S2	2	32.37	31.87	32.37	31.87
S2	4	14.40	14.20	14.83	14.40
S2	6	16.78	16.49	7.77	7.43
S2	8	5.38	5.30	4.58	4.53
S2	10	5.55	5.39	3.35	3.29
S2	12	0.00	0.00	0.00	0.00
S3	2	10.17	10.01	10.17	10.01
S3	4	6.77	6.66	2.33	2.28
S3	6	2.93	2.85	1.21	1.18

Scenario	K	Parameter grouping	Parameter test	Response grouping	Response test
S3	8	1.02	1.01	1.12	1.10
S3	10	0.92	0.89	0.75	0.74
S3	12	0.00	0.00	0.00	0.00

S4.2 PCC trajectories, offsets and internal voltages

Table S5 reports both independent inputs and retains the initial PCC active-power offsets. A small dynamic error does not remove these offsets. The table includes the exact 12-group structural equivalent, whose small production-run differences arise at the solver tolerance. The stricter paired control reduces its relative error to between $1.2 \cdot 10^{-6}$ and $8.7 \cdot 10^{-6}$ in the saved validation summary.

Table S5: PCC Errors, Including Initial Active-Power Offset

Scenario	Model	Input	P_0 offset MW	P RMS error MW	P relative %	Q relative %
S1	K12	grouping	0.00000	0.00001	0.000	0.001
S1	K12	test	0.00000	0.00088	0.063	0.070
S1	K4_param	grouping	-0.00565	0.03663	1.336	1.638
S1	K4_param	test	-0.00584	0.02761	1.983	2.449
S1	K4_resp	grouping	-0.00516	0.02235	0.815	1.050
S1	K4_resp	test	-0.00532	0.01643	1.180	1.541
S1	K2	grouping	-0.00626	0.04759	1.736	2.175
S1	K2	test	-0.00648	0.03589	2.579	3.260
S2	K12	grouping	0.00000	0.00011	0.011	0.011
S2	K12	test	0.00000	0.00017	0.020	0.022
S2	K8_param	grouping	-0.00285	0.01031	1.061	1.800
S2	K8_param	test	-0.00311	0.01029	1.216	1.820
S2	K8_resp	grouping	-0.00362	0.01004	1.033	1.625
S2	K8_resp	test	-0.00306	0.01026	1.212	1.870
S2	K6_param	grouping	-0.00270	0.02835	2.919	3.677
S2	K6_param	test	-0.00387	0.02893	3.417	4.120
S2	K6_resp	grouping	-0.00477	0.01381	1.422	2.304
S2	K6_resp	test	-0.00490	0.01335	1.577	2.291
S3	K12	grouping	0.00000	0.00000	0.000	0.000
S3	K12	test	0.00000	0.00003	0.002	0.002
S3	K4_param	grouping	-0.00625	0.01921	1.595	2.066
S3	K4_param	test	-0.00596	0.02187	1.094	1.473
S3	K4_resp	grouping	-0.00593	0.00944	0.784	1.103
S3	K4_resp	test	-0.00493	0.00991	0.496	0.763
S3	K2	grouping	-0.00691	0.02752	2.286	2.973
S3	K2	test	-0.00659	0.03160	1.581	2.080

The local voltage response is a more demanding target than the common port. Fig. S1 compares the preselected representative chain of S1 with the equivalent containing its hall. This chain has the largest full-model voltage excursion. Parameter grouping combines its hall with halls 2 and 5, whereas response grouping keeps it separate. The parameter-feature equivalent gives a 44.85% dynamic RMS error for this voltage, even though its PCC active-power error is 1.34%. The response-feature voltage error is 0.15% because this hall remains a separate group. This accuracy does not establish preservation of the voltages in its other, merged groups. Across the recorded merged-hall comparisons, local voltage errors range from approximately 45% to 69%. These errors describe differences relative to the voltage change from equilibrium. They are not a

45% to 69% deviation of the absolute operating voltage. A model accepted for PCC response must not be used to infer the voltage stress of an individual merged chain.

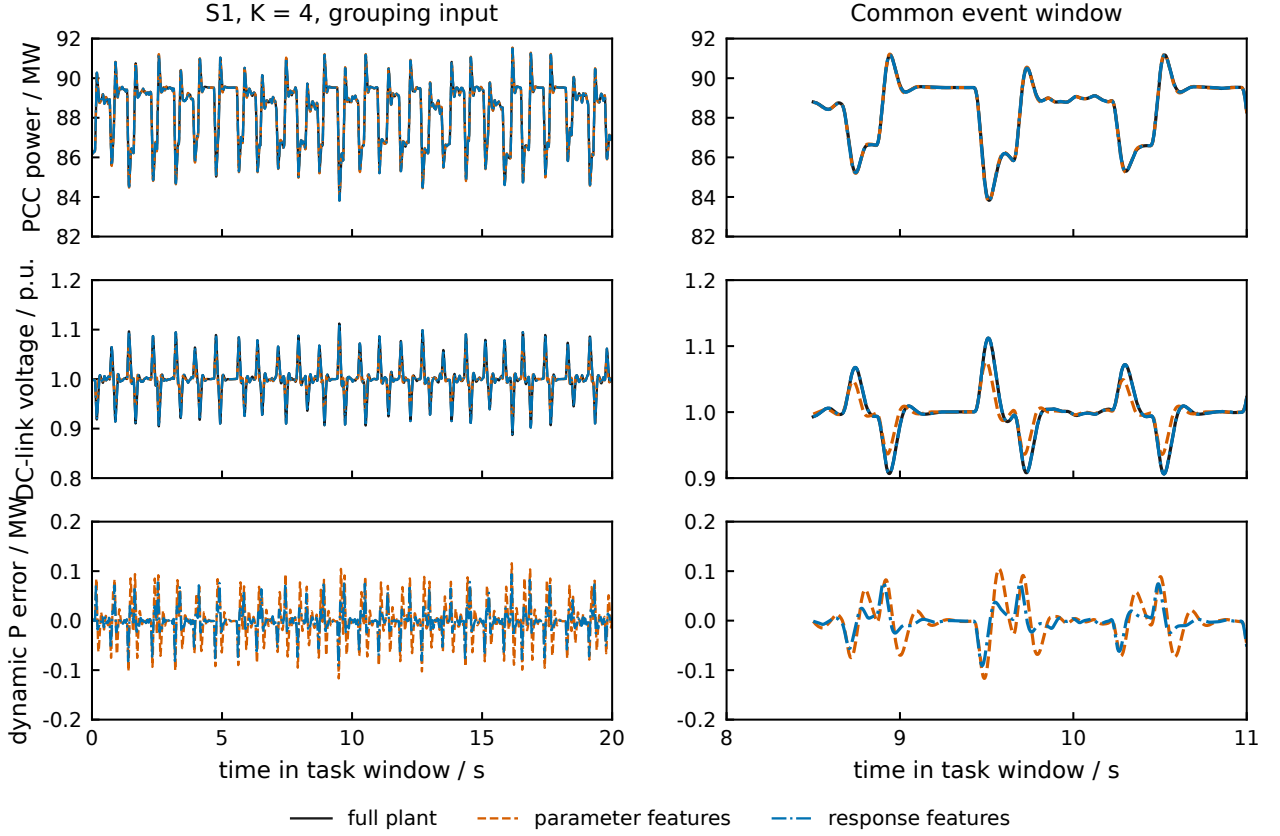


Figure S1: S1 with four groups at the grouping input. A close PCC trajectory coexists with an inaccurate internal DC-link response. Both columns use the same models, with an overview and a common event window. The equivalent voltage is that of the chain containing the selected full-model hall.

Increasing the model order can also reduce the difference between grouping methods. Fig. S2 shows the S2 eight-group case. Its PCC active-power dynamic errors are 1.06% for parameter features and 1.03% for response features at the grouping input. The corresponding test values are 1.22% and 1.21%. This case supports comparable time-domain accuracy, rather than a substantial advantage of either method. The frequency-domain criterion still distinguishes the two, since their grouping-input band errors are 5.38% and 4.58%.

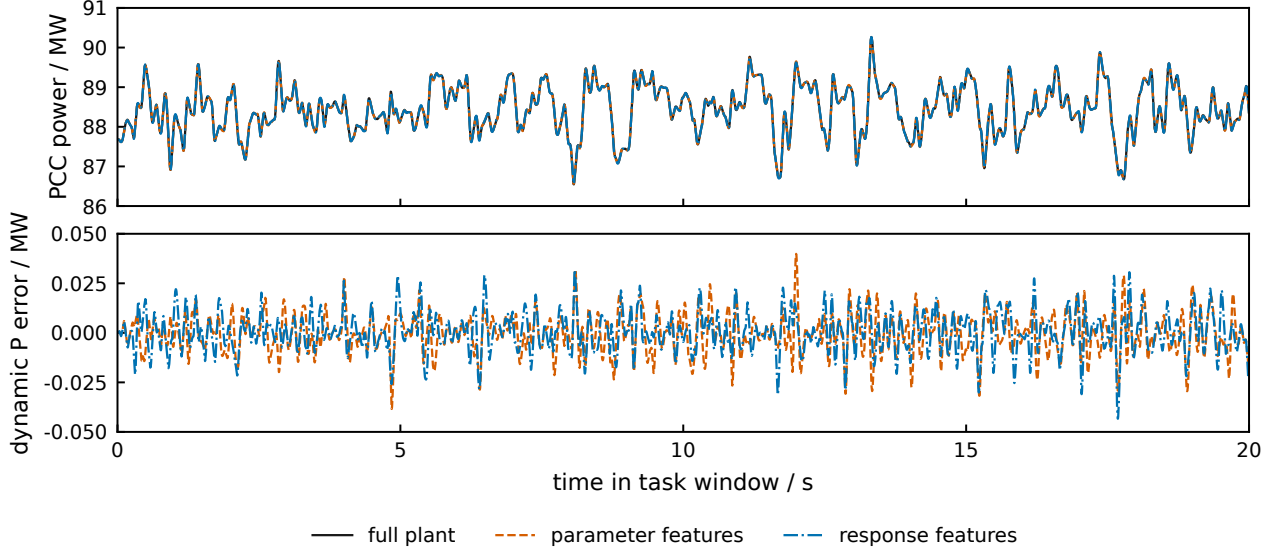


Figure S2: S2 with eight groups at the grouping input. Both feature sets give similar PCC active-power trajectories and dynamic errors.

S4.3 Numerical checks and cost interpretation

All six full-model operating points are stable after excluding the common-angle zero mode. The largest remaining real part lies near -0.117 s^{-1} . At the grouping operating points, the full model has 1873 equations and variables, with initial residual at most $5.8 \cdot 10^{-12}$. A two-second stationary check gives drift at most $4.5 \cdot 10^{-13}$. The 12-group port response differs from the full response by at most $9.3 \cdot 10^{-13}$ in the small-signal comparison.

The six phase-input full-model runs keep the recorded converter-current measure below 0.689 p.u., the modulation ratio below 1.142 and the DC-link voltage between 0.876 and 1.113 p.u. These checks establish the reference model’s behavior for the selected inputs. They do not validate converter protection or establish field operating limits.

Serial integration of 61 s takes median times of 17.56, 38.62 and 16.71 s for the full model in S1, S2 and S3. The exact hall equivalents take 12.76, 28.94 and 12.19 s. The smaller models in the main table take approximately 10 to 27 s. The build and integration records, including the three timing rounds, are retained in `data/p3_summary.json`. Model size falls more than runtime because phase events and 0.01 s input knots remain in every model. The reported timing therefore supports an implementation-specific reduction by a factor of 1.3 to 1.6.

S4.4 Evidence limits

The fast inputs combine literature timing with measured node-power levels. They are constructed scenarios. They do not constitute measured PCC transients, measured synchronization across an operating facility, or a high-bandwidth measurement of the NLR training cycles. Inference remains at a constant request rate within each task window.

The main comparisons use one electrical design. Earlier random-design experiments use different, slower input conditions and remain separate historical evidence. They do not establish the same error values for the phase inputs. The quasi-static network represents electromagnetic line behavior only through fixed-frequency impedances. A network with differential electromagnetic states is required to test the resulting approximation across the full 20 Hz band. The supply-chain model excludes switching ripple, device limiting, protection and battery-only operation. The present conclusions concern task-driven PCC quantities under the stated conditions.

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